

Lecture 6 Quiz Questions and Answers

1

Select the correct statement/s related to PD control

steady state error of a second-order system with proportional plus derivative feedback control is identical to first order system with proportional plus derivative control

☐ 9%

proportional control increase the frequency of the response

☐ 6%

by tuning it is possible to eliminate oscillations from the response

☐ 12%

1 and 2

☐ 5

all of the above ☒

☒ 21%

2

Select correct statement/s related to integral control

first order system with integral control action eliminates the steady state error ☒

☒ 3%

higher K_i (say 20) gives a highly damped oscillation for first order systems

☐ 23%

lower K_i (say 0.2) gives fast exponential response for first order systems

☐ 5%

none of the above

☐ 8%

1 and 3

☐

3

Select the correct statement/s related to PI control (first-order)

integral gain influence the frequency of the response

☐ 17%

proportional gain used to achieve the required damping

☐ 21%

addition of proportional control to integral control action has no effect on the steady state error

☐ 17%

1 and 3

☐all of the above ☒☐ 17%

4

Select the correct statement/s related to PID control (first order)

48

addition of derivative control to proportional and integral control action has no effect on the steady state error

☐ 0%

controller may be tuned

☐ 17%

actual response of the system doesn't dependent on damping ratio

☐ 0%1 and 2 ☒☐ 52

2 and 3

☐ 31%

